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Supporting Information

Non-Monotonic Computational Complexity in Lignin Assembly Reveals a Hardwood Transition Peak Across 400 Million Years of Plant Evolution — Supporting Information

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Abstract

This document provides Supporting Information for the main text: theoretical background on Kolmogorov complexity and the $t_{\mathbf{gen}}$ proxy (Section S1), complete statistical tables for both experiments (Section S2), the LGS generation machinery and combinatorial basis of the hardwood complexity peak (Section S3), parameter grounding and experimental configurations (Section S4), and execution environment and reproducibility details, including a sequential re-run validating Experiment 2 (Section S5).

Contents

S1. Kolmogorov Complexity and the $t_{\mathbf{gen}}$ Proxy

S1.1. Kolmogorov Complexity: Definition and Properties

Definition S1.1 — Kolmogorov Complexity

Let U be a universal Turing machine (UTM), x a binary string, and p a program (binary string). The **Kolmogorov complexity** of x is:

$$K_U(x) = \min_{p\,:\,U(p)=x} \ell(p)$$

where $\ell(p)$ is the length of program p in bits.

$K(x)$ is the length of the *shortest computer program* that generates x and halts. It measures the minimum information content intrinsic to x: a string with internal regularities (repetitions, patterns) has low K; a truly random string has $K(x) \approx \ell(x)$.

Uncomputability. $K(x)$ is uncomputable as a consequence of the Halting Problem. No algorithm can compute $K(x)$ exactly for arbitrary x. Practical applications therefore rely on approximations: lossless compression (upper bound), the Coding Theorem Method (CTM; Delahaye & Zenil 2012) applicable to short strings, or the Block Decomposition Method (BDM) for longer objects.

Invariance Theorem. For any two universal Turing machines U and V:

$$|K_U(x) - K_V(x)| \le c_{UV}$$

where c_{UV} is a machine-dependent constant independent of x. This makes $K(x)$ an intrinsic property of the string, not an artifact of the computational model.

S1.2. Algorithmic Probability and the Coding Theorem

Definition S1.2 — Algorithmic Probability (Solomonoff, 1964)

$$P_U(x) \;=\; \sum_{\substack{p\,:\,U(p)=x\\ \text{halts}}} 2^{-\ell(p)}$$

The probability that a randomly drawn binary program causes U to output x.

Theorem S1.1 — Coding Theorem (Levin, 1974)

$$\log_2 P_U(x) \;\ge\; -K_U(x) - O(1) \quad\Leftrightarrow\quad \boxed{P_U(x) \;\ge\; 2^{-K_U(x)-c}}$$

Reading the equation. Lower $K(x)$ (shorter description) implies higher $P(x)$ (greater probability of arising by random process). The bound is exponential: reducing $K(x)$ by n bits multiplies $P(x)$ by at least 2^n.

Evolutionary implication (Johnston *et al.* PNAS 2022): phenotypes with lower K are exponentially overrepresented in biological genotype–phenotype maps. This "simplicity bias" shapes evolutionary trajectories independently of natural selection ("arrival of the frequent", Wagner 2014).

S1.3. The $t_{\mathbf{gen}}$ Proxy: Formal Relationship

The LGS algorithm is a rejection sampler. Let $P_{\mathrm{LGS}}(S \mid \theta)$ be the probability that one draw from the LGS prior produces a valid structure S given parameters $\theta = (f_{\beta\text{-O-4}}, r_{\mathrm{S/G}}, \mathrm{DP})$. Then the expected number of iterations before acceptance is:

$$R(S;\theta) \;=\; \frac{1}{P_{\mathrm{LGS}}(S \mid \theta)}$$

and wall-clock time is:

$$t_{\mathrm{gen}}(S;\theta) \;\approx\; t_{\mathrm{step}} \cdot R(S;\theta) + t_{\mathrm{overhead}} \;\propto\; \frac{1}{P_{\mathrm{LGS}}(S \mid \theta)}$$

If the LGS prior approximates the chemical probability distribution of lignin polymerization:

$$P_{\mathrm{LGS}}(S \mid \theta) \;\approx\; 2^{-\tilde{K}(S|\theta)}$$

then $t_{\mathrm{gen}} \propto 2^{\tilde{K}(S|\theta)}$ — an exponential relationship between generation time and conditional descriptional complexity.

S1.4. Comparison: $K(x)$ vs. $t_{\mathbf{gen}}$

Key point

The proxy is empirically justified, not formally derived. We do not claim $t_{\mathrm{gen}} = K(x)$ or that t_{gen} satisfies the Invariance Theorem. We claim that t_{gen} is a computable, reproducible measure that captures domain-specific conditional complexity, and test empirically whether this quantity predicts evolutionary trends in lignin.

Table S1 Comparison between Kolmogorov complexity $K(x)$ and the t_{gen} proxy used in this work.

Property	$K(x)$	t_{gen}
Computability	Uncomputable	Directly measurable
Universality	UTM-invariant (up to additive constant)	LGS-specific; not universal
Formal connection to $P(x)$	Coding Theorem: $P(x) \geq 2^{-K(x)-c}$	Rejection rate: $t_{\text{gen}} \propto P_{\text{LGS}}^{-1}$
What it measures	Intrinsic complexity of x	Conditional complexity given LGS prior θ
String length	Applicable to all binary strings	Applied here: SMILES of DP$\in [5, 10]$
Algorithm invariance	Yes (Invariance Theorem)	No — t_{gen} is LGS-specific
Evolutionary claim	$K(x) \downarrow \Rightarrow P(x) \uparrow$ (Coding Theorem)	$t_{\text{gen}} \downarrow \Rightarrow P_{\text{LGS}} \uparrow \Rightarrow$ *possibly* favored
Validation in biology	Johnston *et al.* PNAS 2022 (proteins, RNA)	This work (lignin, first application)

Table S2 Two-way ANOVA for Experiment 1 (t_{gen} as dependent variable; $n = 270$, residual $df = 261$). SS = sum of squares; MS = mean square; η_p^2 = partial eta-squared.

Source	**SS**	df	**MS**	F	η_p^2
Plant type (group)	8406.7	2	4203.4	1342.8	0.911
DP	66383.1	2	33191.6	10603.2	0.988
Plant type × DP	26503.0	4	6625.7	2116.6	0.970
Residual	817.0	261	3.13		

All $p < 0.001$ (plant type: $p = 4.2 \times 10^{-138}$; DP: $p = 1.2 \times 10^{-250}$; interaction: $p = 1.6 \times 10^{-197}$).

Table S3 Tukey HSD pairwise comparisons for Experiment 1, factor: plant type (pooled over DP levels). Hedges' g used as effect size.

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ	g	p_{Tukey}
GR	HW	19.64	30.00	−10.4	−0.473	< 0.001
HW	SW	30.00	17.11	+12.9	+0.615	< 0.001
GR	SW	19.64	17.11	+2.5	+0.213	0.636 (ns)

Table S4 Tukey HSD pairwise comparisons for Experiment 1, factor: DP (pooled over plant types).

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ	g	p_{Tukey}
DP=10	DP=5	42.34	4.07	+38.3	+2.760	< 0.001
DP=10	DP=8	42.34	20.34	+22.0	+1.547	< 0.001
DP=5	DP=8	4.07	20.34	−16.3	−5.023	< 0.001

S2. Complete Statistical Analysis

S2.1. Experiment 1: Full ANOVA Table

S2.2. Experiment 1: Tukey HSD Post-Hoc Tests

S2.3. Experiment 1: Normality and Homoscedasticity

S2.4. Experiment 2: Full ANOVA Table

S2.5. Experiment 2: Per-Cell Descriptive Statistics

S2.6. Experiment 2: Tukey HSD Post-Hoc Tests

S2.7. Non-Parametric Robustness Check

S2.8. Post-Hoc Power Analysis

S3. LGS Generation Machinery and the HW Complexity Peak

S3.1. The Four-Stage Generation Pipeline

The LGS algorithm (Eswaran *et al.* 2022) operates in four stages documented in their Methods section:

Stage 1 — Monomer sequencing (Heap's algorithm). All unique permutations of G and S monomers are generated for the given S/G ratio and DP. The number of unique sequences is:

$$N_{\text{seq}} = \binom{n}{k}$$

where n = DP and k = number of S units, determined by $k = \lfloor n \cdot r_{\text{S/G}}/(1 + r_{\text{S/G}}) \rceil$.

Stage 2 — Conditional linkage generation. For each monomer sequence, bond types are sampled conditionally on

Table S5 Shapiro–Wilk normality test per cell for Experiment 1 ($n = 30$ per cell). CV = coefficient of variation (%). * indicates $p < 0.05$ (mild non-normality); all ANOVA results confirmed by non-parametric Kruskal–Wallis tests.

Group	DP	W	p	CV (%)
SW	5	0.944	0.118	6.3
SW	8	0.910	0.015*	8.2
SW	10	0.954	0.222	5.1
HW	5	0.951	0.174	6.4
HW	8	0.975	0.693	11.4
HW	10	0.977	0.750	4.8
GR	5	0.918	0.024*	5.4
GR	8	0.906	0.012*	8.4
GR	10	0.953	0.206	6.1

Table S6 Two-way ANOVA for Experiment 2 (t_{gen} as dependent variable; $n = 239$, residual $df = 230$).

Source	SS	df	MS	F	η_p^2
$f_{\beta\text{-O-4}}$	1,540,810	2	770,405	284.6	0.712
S/G ratio	351,765	2	175,883	65.0	0.361
$f_{\beta\text{-O-4}}$×S/G	210,995	4	52,749	19.5	0.253
Residual	622,613	230	2,707		

All $p < 0.001$ ($f_{\beta\text{-O-4}}$: $p = 6.2 \times 10^{-63}$; S/G: $p = 4.3 \times 10^{-23}$; interaction: $p = 8.0 \times 10^{-14}$).

Table S7 Per-cell descriptive statistics for Experiment 2. Mean and SD of t_{gen} (seconds). CV = coefficient of variation (%). n = number of successful runs (of 30 planned).

$f_{\beta\text{-O-4}}$	S/G	n	Mean (s)	SD (s)	CV (%)
0.54 (SW)	0.0	27	100.83	42.52	42.2
0.54 (SW)	0.8	28	116.45	18.48	15.9
0.54 (SW)	1.5	27	108.46	27.53	25.4
0.69 (HW)	0.0	24	185.37	35.54	19.2
0.69 (HW)	0.8	28	374.30	82.79	22.1
0.69 (HW)	1.5	27	288.86	90.60	31.4
0.77 (GR)	0.0	27	72.46	17.80	24.6
0.77 (GR)	0.8	25	152.32	24.80	16.3
0.77 (GR)	1.5	26	140.60	62.01	44.1

Table S8 Tukey HSD and Cohen's d for Experiment 2, factor: $f_{\beta\text{-O-4}}$ (pooled over S/G levels).

A	B	$\bar{t}_A$	$\bar{t}_B$	Δ (s)	d	p_{Tukey}
HW (0.69)	SW (0.54)	287.7	108.7	+179.0	+2.28	< 0.001
HW (0.69)	GR (0.77)	287.7	120.8	+166.9	+1.97	< 0.001
GR (0.77)	SW (0.54)	120.8	108.7	+12.1	+0.28	0.528 (ns)

local monolignol identity. Table S13 reproduces the coupling rules from Table 3 of Eswaran *et al.* (2022).

Stage 3 — Rejection sampling. Candidate structures are accepted only if they satisfy the `bondconfig` frequency targets. Structures that fail are discarded and re-drawn. This is the stage where t_{gen} varies.

Stage 4 — SMILES output. Accepted structures are converted to canonical SMILES via the Chemistry Development Kit (CDK).

Table S9 Two-sample t-tests restricted to the *natural plant-type configurations* (canonical cells: SW/SW, HW/HW, GR/GR; not pooled over S/G). These are the biologically meaningful comparisons discussed in the main text (Section 5.2). Unlike the pooled comparison above, which mixes in S/G levels that do not correspond to any natural plant type, this comparison shows that SW/SW and GR/GR are *not* statistically indistinguishable.

A	**B**	n_A	n_B	$\bar{t}_A$	$\bar{t}_B$	Δ (s) / d	$t(df)$, p
HW/HW	SW/SW	27	27	288.9	100.8	+188.0 / +2.66	$t(52) = 9.76$, $p = 2.4 \times 10^{-13}$
HW/HW	GR/GR	27	25	288.9	152.3	+136.5 / +2.02	$t(50) = 7.28$, $p = 2.2 \times 10^{-9}$
GR/GR	SW/SW	25	27	152.3	100.8	+51.5 / +1.47	$t(50) = 5.28$, $p = 2.8 \times 10^{-6}$

Table S10 Tukey HSD and Cohen's d for Experiment 2, factor: S/G ratio (pooled over $f_{\beta\text{-O-4}}$ levels).

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ (s)	d	p_{Tukey}
S/G=0.8	S/G=0.0	216.7	117.0	+99.6	+1.00	< 0.001
S/G=1.5	S/G=0.0	179.8	117.0	+62.8	+0.75	< 0.001
S/G=0.8	S/G=1.5	216.7	179.8	+36.9	+0.32	0.053 (ns)

Table S11 Kruskal–Wallis tests for Experiment 2, confirming ANOVA results under no distributional assumptions.

Factor	H	df	p
$f_{\beta\text{-O-4}}$	135.6	2	3.6×10^{-30}
S/G ratio	42.4	2	6.3×10^{-10}

Table S12 Post-hoc statistical power for key comparisons (two-sample t-test, $\alpha = 0.05$, $n = 30$ per group). Power computed from observed effect sizes.

Comparison	d	n	**Power**
Exp. 2: HW vs. SW (pooled)	2.28	30	> 0.9999
Exp. 2: HW vs. GR (pooled)	1.97	30	> 0.9999
Exp. 2: SW vs. GR (pooled)	0.28	30	0.191
Exp. 2: SW/SW vs. GR/GR (canonical)	1.47	25–27	> 0.999
Exp. 1: SW vs. HW at DP=8	1.50	30	> 0.9999

Note: The *pooled* SW vs. GR comparison (which mixes in non-natural S/G levels) is a small effect ($d = 0.28$) that this study was underpowered to detect ($n \approx 200$ would be needed for 80% power). However, the biologically relevant *canonical-cell* comparison, SW/SW vs. GR/GR (Table S9), is a large effect ($d = 1.47$) that is already well-powered at $n = 25$–27 and is statistically significant ($p < 0.001$). The two comparisons answer different questions: the pooled test asks about the $f_{\beta\text{-O-4}}$ main effect; the canonical-cell test asks whether the two natural flanking plant types are equivalent, and the answer is no.

Table S13 Monomer coupling patterns reproduced from Eswaran *et al.* (2022), Table 3. ✓ = bond type possible; × = bond type forbidden. S units lack the free C-5 position of G units, preventing β-5 and 5-5 C–C bonds.

		Linear			**Branched**	
M1	**M2**	β-O-4	β-β	β-5	4-O-5	5-5
G	G	✓	✓	✓	✓	✓
S	S	✓	✓	×	×	×
S	G	✓	✓	✓	×	×
G	S	✓	✓	×	✓	×

S3.2. Why the HW Complexity Peak Is Not an Artifact

The combinatorial argument

Table S14 Combinatorial analysis of sequence spaces and bond constraints at DP = 10 for the three plant types. "C–C req." = fraction of non-β-O-4 bonds in `bondconfig` (approx. $1 - f_{\beta\text{-O-4}}$ accounting for all C–C and mixed bond types).

Config	$\binom{10}{k}$	C–C req.	$\bar{t}_{\mathrm{gen}}$ (s)	Bottleneck
SW (S/G= 0)	$\binom{10}{0} = 1$	46%	100.8	G-only: no conditional constraint
HW (S/G= 1.5)	$\binom{10}{6} = 210$	31%	288.9	C–C bonds require G positions in G+S pool
GR (S/G= 0.8)	$\binom{10}{4} = 210$	23%	152.3	High β-O-4 substantially relaxes the G-position constraint (but does not eliminate it)

At DP = 10, HW ($r_{\mathrm{S/G}} = 1.5$) and GR ($r_{\mathrm{S/G}} = 0.8$) have *identical* sequence spaces:

$$N_{\mathrm{seq}}(\mathrm{HW}) = \binom{10}{6} = 210 \qquad N_{\mathrm{seq}}(\mathrm{GR}) = \binom{10}{4} = 210$$

Yet HW generates 1.9× slower (288.9 s vs. 152.3 s, natural plant-type cells). The difference cannot be attributed to enumeration size. It arises from the bond constraint: HW requires 31% C–C bonds in its `bondconfig`, but C–C bonds are restricted to G-unit positions (Table S13). With 6 S units and 4 G units in the HW pool, the algorithm must repeatedly find G-unit positions for C–C bonds while placing S units elsewhere — a combinatorially constrained search. GR requires only 23% C–C bonds and has 77% β-O-4 (accessible to both G and S), which dissolves the constraint.

This argument demonstrates that the HW complexity peak is a consequence of the *chemically correct* encoding of monolignol bonding restrictions in LGS (Eswaran *et al.* 2022, Table 3), not of any hardcoded optimization or tuning of the algorithm for specific $f_{\beta\text{-O-4}}$ values.

The same logic explains why GR/GR (152.3 s) does not fully revert to the SW/SW level (100.8 s), consistent with the significant residual difference reported in the main text (Section 5.2): GR still draws 23% of its bonds from the G-constrained C–C pool, whereas SW's all-G composition never encounters the constraint at all, regardless of its higher nominal C–C requirement (46%). GR reduces, but does not eliminate, the positional bottleneck that peaks at HW.

Table S15 Native milled wood lignin (MWL) interunit linkage frequencies (% of total bonds) used to define parameter bounds for LGS configuration. Values represent experimental data from primary MWL studies. Central values (midpoint of range) were used in all experiments and normalized to sum to 100 for the `bondconfig` weight format.

Plant type	β-O-4	β-5	β-β	4-O-5	5-5
Softwood (SW)	43–50	9–12	2–4	4–8	18–25
Hardwood (HW)	50–65	4–8	3–8	5–8	5–12
Grass (GR)	65–80	3–7	3–6	5–10	3–8

Sources: Eswaran *et al.* (2022); Obrzut *et al.* (2023); Lourenço & Pereira (2018); Chakar & Ragauskas (2004).

Table S16 Complete `bondconfig` parameters (normalized to sum to 100) for all experimental configurations. Experiment 1 configurations are identified by plant type and DP; Experiment 2 configurations by $f_{\beta\text{-O-4}}$ source and S/G level. All Experiment 2 runs at DP = 10.

Exp.	**Config**	**DP**	f_β	S/G	BO4	B5	BB	55
1	SW_DP5	5	0.54	0.0	54	11	3	25
1	SW_DP8	8	0.54	0.0	54	11	3	25
1	SW_DP10	10	0.54	0.0	54	11	3	25
1	HW_DP5	5	0.69	1.5	69	7	7	10
1	HW_DP8	8	0.69	1.5	69	7	7	10
1	HW_DP10	10	0.69	1.5	69	7	7	10
1	GR_DP5	5	0.77	0.8	77	5	4	6
1	GR_DP8	8	0.77	0.8	77	5	4	6
1	GR_DP10	10	0.77	0.8	77	5	4	6
2	SW_fBO4_sg0	10	0.54	0.0	54	11	3	25
2	SW_fBO4_sg08	10	0.54	0.8	54	11	3	25
2	SW_fBO4_sg15	10	0.54	1.5	54	11	3	25
2	HW_fBO4_sg0	10	0.69	0.0	69	7	7	10
2	HW_fBO4_sg08	10	0.69	0.8	69	7	7	10
2	HW_fBO4_sg15	10	0.69	1.5	69	7	7	10
2	GR_fBO4_sg0	10	0.77	0.0	77	5	4	6
2	GR_fBO4_sg08	10	0.77	0.8	77	5	4	6
2	GR_fBO4_sg15	10	0.77	1.5	77	5	4	6

S4. Parameter Grounding and Experimental Configurations

S4.1. Native MWL Literature Ranges

S4.2. Normalized LGS Configurations for All 18 Cells

S5. Execution Environment and Reproducibility

S5.1. Hardware and Software

S5.2. Sequential Execution (Experiment 1) vs. Parallel Sandboxing (Experiment 2)

The two experiments used different execution strategies, and this difference has direct consequences for the comparability of absolute t_{gen} values across experiments.

Experiment 1 was executed as a single sequential process: LGS runs were submitted one at a time, with the next run launched only after the previous one completed. Each measured t_{gen} therefore reflects an isolated LGS process with exclusive access to the machine's resources.

Experiment 2 was executed using a parallel sandboxing strategy intended to reduce total runtime. LGS reads from a fixed `resources/project_config.yaml` path and writes to fixed `output/` subdirectories relative to the working directory. To enable parallel execution across 8 workers without file-system conflicts, 8 isolated working directories (`worker_0` through `worker_7`) were created, each containing:

- A symbolic link to the shared `lgs_gen.jar`
- Independent `resources/` subdirectory with the cell-specific `project_config.yaml`
- Independent `output/` subdirectory for LGS output

Table S17 Execution environment for all LGS runs.

Component	Version / Specification
Operating system	Ubuntu 24.04 LTS
Java runtime	OpenJDK 17 (LGS is a Java process)
Python	3.13
Statistical analysis	`scipy` 1.13, `pingouin` 0.5, `pandas` 2.2
Execution mode, Exp. 1	Sequential (one LGS process at a time)
Execution mode, Exp. 2	Parallel, 8 nominal workers (`ThreadPoolExecutor`); per-task duration inflated 3.8–4.5× relative to uncontended execution (directly measured, Section S5.3)
Timing measurement	`time.time()` bracketing `subprocess.run()`
LGS version	`lgs_gen.jar` (Eswaran *et al.* 2022)
LGS source code	`github.com/sudhacheran/lignin-structure-generator`

Jobs were distributed across workers via `ThreadPoolExecutor (max_workers=8)` with randomized task order to prevent systematic load imbalance. Output files were deleted immediately after t_{gen} was recorded to manage disk usage.

This parallelization did not achieve full 8× throughput. Rather than infer this indirectly from inter-completion timestamp spacing, Section S5.3 reports a direct measurement: a sequential re-run of the complete Experiment 2 grid shows that per-task t_{gen} under the 8-worker parallel regime is inflated by a factor of 3.8–4.5× (mean ≈ 4.1×), consistently across all 9 cells, relative to the same task executed alone. This indicates substantial resource contention among simultaneously running JVM processes (e.g. shared physical cores behind 8 logical threads, shared memory bandwidth, and garbage collection pauses) — realized throughput is far below the nominal 8-worker capacity. This contention is the most plausible explanation for two features of the Experiment 2 data reported in the main text: the substantially higher per-cell coefficient of variation relative to Experiment 1 (up to 44%, Table S7, vs. <12% in Table S5) and the 31 runs that failed under JVM memory constraints (Section 4.2 of the main text).

Consequence for interpretation. Because Experiment 1 and Experiment 2 were executed under different concurrency regimes, their absolute t_{gen} values are not directly comparable: the same configuration (HW at DP = 10, $f_{\beta\text{-O-4}}$ = 0.69, S/G = 1.5) yields 68.1 ± 3.3 s under Experiment 1's sequential execution but 288.9 ± 90.6 s under Experiment 2's contended parallel execution — a ∼4× difference attributable to execution environment rather than to chemistry. All hypothesis tests in this paper remain valid, because every comparison is made *within* a single experiment, where all cells were collected under the same execution regime and in randomized task order: the Experiment 1 gradient test compares plant types measured purely sequentially, and the Experiment 2 factorial test compares $f_{\beta\text{-O-4}}$ and S/G levels measured purely under the same 8-worker parallel regime. A remaining concern is whether contention itself contributes to the *shape* of the Experiment 2 result: slower cells remain active longer and are thereby exposed to more concurrent-sibling contention than faster cells, which could in principle amplify (though not create from nothing) the relative gap between the hardwood cell and the other two. Section S5.3 reports a full sequential re-run of the Experiment 2 grid that directly addresses this concern.

S5.3. Sequential Validation of the Experiment 2 Result

To test whether the parallel execution regime of Experiment 2 inflates or distorts the non-monotonic pattern, the complete 3 × 3 factorial grid (9 configurations × 30 replicates = 270 runs) was re-executed using Experiment 1's sequential protocol — one LGS process at a time, no `ThreadPoolExecutor`, no worker sandboxing — via a standalone script (`sequential_rerun/run_sequential.py` in the code repository). 269 of 270 runs completed successfully (one run was interrupted by an unrelated process-management event during the rerun and is excluded, mirroring how the 31 JVM-memory failures in the original parallel run were excluded).

Two findings from the sequential re-run

(1) The contention inflation factor is uniform, not differential. The parallel/sequential ratio is 3.83–4.47× across all 9 cells, with no relationship to how slow the cell is (the slowest cell, HW at S/G= 0.8, has ratio 3.87×; two of the fastest cells have the *highest* ratios). This directly answers the concern raised above: contention does not preferentially amplify the hardwood cells. The non-monotonic peak at HW/HW is present in the uncontended data (SW/SW = 22.9 s, GR/GR = 39.7 s, HW/HW = 74.7 s) and is not an artifact of parallel execution.

(2) The SW/SW vs. GR/GR asymmetry also replicates independently. A two-way ANOVA on the sequential data confirms both main effects and their interaction ($f_{\beta\text{-O-4}}$: F = 755.0, η^2 = 0.853, $p = 5.1 \times 10^{-109}$; S/G: F = 256.1, η^2 = 0.663, $p = 3.5 \times 10^{-62}$; interaction: F = 64.2, η^2 = 0.497, $p = 1.1 \times 10^{-37}$). At the canonical cells, GR/GR (39.7 s) is significantly slower than SW/SW (22.9 s; d = 2.46, $t(58)$ = 9.52, $p = 1.8 \times 10^{-13}$), and both are far below HW/HW (74.7 s; vs. SW/SW: d = 6.96, $p = 6.1 \times 10^{-34}$; vs. GR/GR: d = 3.98, $p = 7.6 \times 10^{-22}$). This independently reproduces,

Table S18 Sequential re-run (uncontended) vs. original parallel means for all 9 cells of the Experiment 2 grid. Ratio = parallel mean / sequential mean.

$f_{\beta\text{-O-4}}$	S/G	n_{seq}	Seq. mean (s)	Par. mean (s)	**Ratio**
0.54 (SW)	0.0	30	22.92	100.83	4.40
0.54 (SW)	0.8	30	29.95	116.45	3.89
0.54 (SW)	1.5	30	25.22	108.46	4.30
0.69 (HW)	0.0	30	41.43	185.37	4.47
0.69 (HW)	0.8	30	96.79	374.30	3.87
0.69 (HW)	1.5	29	74.72	288.86	3.87
0.77 (GR)	0.0	30	16.37	72.46	4.43
0.77 (GR)	0.8	30	39.74	152.32	3.83
0.77 (GR)	1.5	30	34.02	140.60	4.13

in a freshly collected uncontended dataset, the same asymmetric pattern reported for the parallel data in the main text (Section 5.2) and Table S9: grasses do not fully revert to the ancestral softwood complexity level.

Taken together, the sequential re-run resolves the concern raised earlier in this section: the non-monotonic HW peak and the partial (not complete) SW/GR asymmetry are both properties of the LGS assembly problem itself, replicated under a contention-free execution regime, not artifacts of the parallel sandboxing strategy used to collect the original Experiment 2 data.

S5.4. Data and Code Availability

All configuration files, analysis notebooks, and raw results are publicly available at:

`https://github.com/In-Silico-RG/LSG_Molec_Evol`

Zenodo archive (DOI-minted, version-stable):
`doi:10.5281/zenodo.21195547`

Repository contents:

- `configs/experiment1/` — 9 YAML files (SW/HW/GR × DP5/8/10)
- `configs/experiment2/` — 9 YAML files ($f_{\beta\text{-O-4}}\times$ S/G orthogonal)
- `notebooks/03-Run_Evolutionary_Experiment.ipynb` — Experiment 1 execution notebook (sequential, one `lgs_gen.jar` process at a time)
- `notebooks/05-Parallel_Experiment.ipynb` — Experiment 2 execution notebook (8-worker parallel, auto-generates YAMLs, no external files needed beyond `lgs_gen.jar`)
- `results/evolutionary_experiment_results.csv` — raw data, Experiment 1 (270 rows)
- `results/exp2_parallel_results.csv` — raw data, Experiment 2 (244 rows, 239 successful)
- `sequential_rerun/run_sequential.py` — script for the sequential validation of Experiment 2 (Section S5.3)
- `sequential_rerun/exp2_sequential_results.csv` — raw data for the sequential validation of Experiment 2 (270 rows, 269 successful)
- `figures/` — source files for Figures 1–2 and the graphical abstract (`exp1_barplot`, `exp2_heatmap`, `graphical_abstract`; PDF and PNG)